Chapter 23 worked solutions

Vision–Language and Multimodal Models

Exercises 1–4 work through similarity, retrieval, and a visual bridge by hand. Exercises 9–11 design tests for retrieval on new scenes and for answers that follow the image.

1. Normalize two modalities

We want to rank two texts for the first image by how closely their vectors point in the same direction. First divide each row by its own length. For a two-coordinate row (x, y), the length is $\sqrt{x^2 + y^2}$. The image row lengths are 3 and 2. The text row lengths are $\sqrt{2}$ and 2. With $\mathbf{i}_a$ for image a and $\mathbf{t}_b$ for text b, the unit vectors are

$$\begin{aligned} \mathbf{i}_1 &= (1, 0), & \mathbf{i}_2 &= (0, 1), \\ \mathbf{t}_1 &= (1/\sqrt{2}\ , 1/\sqrt{2}\ ), & \mathbf{t}_2 &= (0, 1). \end{aligned}$$

Now take the dot product of each image with each text. Because the rows have unit length, these dot products are cosine similarities. For the first image, the dot product selects the first coordinate of each text, giving $1/\sqrt{2}$ for text 1 and 0 for text 2. The second image selects the second coordinate, giving $1/\sqrt{2}$ and 1. Multiply each similarity by the scale 2 to get the logits, or scores before softmax. Images index the rows and texts index the columns.

$$L = 2\begin{bmatrix} 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 \\ \sqrt{2} & 2 \end{bmatrix}.$$

The first image ranks **text 1 first and text 2 second**, since $\sqrt{2} > 0$ in its row.

2. Read a row and a column

The first image and first text are a training pair. Their shared logit is $L_{11} = 1$. To find the probability assigned to the pair, softmax exponentiates both scores in the comparison and divides the exponentiated score for the pair by their sum.

- **First image queries the texts.** Read row 1, $(1, 0)$. The competing logit comes from text 2, so the paired probability is $e^1/(e^1 + e^0) \approx 0.7311$.
- **First text queries the images.** Read column 1, $(1, 2)$. The competing logit comes from image 2, so the paired probability is $e^1/(e^1 + e^2) \approx 0.2689$.

Both numerators are e^1. In the row, the paired logit beats its competitor. In the column, the paired logit faces a higher-scoring competitor, which makes the denominator larger and the paired probability smaller.

3. Count more than one valid caption

C3 is the first relevant caption to appear, at rank 2. C1 appears at rank 4. Recall at rank k is the number of relevant captions found in the first k results divided by the total number of relevant captions for the query. Here the denominator is always 2. The hit indicator separately records whether the first k results contain at least one relevant caption.

k	First k results	Relevant captions found	Recall at k	Hit?
1	C2	None	$0/2 = 0$	No
2	C2, C3	C3	$1/2 = 0.5$	Yes
4	C2, C3, C4, C1	C3, C1	$2/2 = 1$	Yes

At rank 2, the query already has a hit, but it has recovered only half of its relevant captions.

4. Mix values before projection

The query gives the two image values equal scores, so it reads them with equal weight. Each score of 0 contributes $e^0 = 1$ to the softmax denominator. The weights are $(1/2, 1/2)$, so the attention output is the average of the two value rows.

$$\frac{1}{2}(2, 1) + \frac{1}{2}(0, 3) = (1, 2).$$

The projection takes the two coordinates of this attention output and produces three features. Multiply the row $(1, 2)$ by the given projection matrix, or equivalently take one copy of its first row and two copies of its second row.

$$(1, 2)\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix} = (1, 0, 2) + 2(0, 1, -1) = (1, 2, 0).$$

The projected output has shape 1×3. There is one row because there was one query, and there are three features because the projection has three columns.

9. Keep related pairs together (optional)

Imagine a photographer takes three nearly identical photos of the same bus stop. Each gets five captions. If one photo and its captions go to training while another photo goes to test, the retrieval system may see almost the same image again. A split by caption could leak even more directly by placing descriptions of the very same photo on both sides.

Use the **scene** as the grouping unit. Give each photo a stable ID and each caption the ID of its parent photo. Build scene groups from source information such as recording session or location, and inspect near-duplicate candidates to catch copies, crops, and similar shots that would otherwise cross groups. Merge any scene groups linked by a near-duplicate cluster before assigning splits. Keep a scene's photos and all their captions together. For example, with 100 suitably sized scene groups, assign 70 to training, 15 to validation, and 15 to test. Record the actual numbers of scenes, photos, and captions. Verify that neither a scene ID nor a duplicate photo cluster spans splits. Choose retrieval settings on validation and freeze them for test.

Build the test gallery from held-out photos and the test text pool from their captions. Keep all five captions for each photo. Write relevance judgments before scoring. For a photo query, mark its five captions as relevant. If the task also credits captions from another photo that accurately describe this one, judge and record those links under a common rule. A description of a blue bus may fit two photos, but a description of a blue bus facing the shelter may apply to only one. A shared scene ID alone does not settle that judgment. Use the same relevance links when reversing direction.

For **image to text**, rank all test captions for each photo. If the first three results contain two of the five relevant captions, its recall at rank three is 2/5. For **text to image**, rank all test photos for each caption. If the caption has only its parent photo marked relevant and that photo ranks second, its recall at one is 0 and its recall at three is 1. If more photos were judged relevant, use their count as the denominator for that query.

Report mean per-query recall separately by direction at fixed cutoffs such as 1, 5, and 10. State the gallery sizes, how you break score ties, the number of defined queries, and any queries excluded for having no relevance labels. An any-hit measure can also be reported separately. The photo with two correct captions at rank three has an any-hit score of 1 and recall of 2/5. If reporting uncertainty, resample by scene so that correlated photos and their captions remain together.

10. Test a spatial relation (optional)

Ask the same question of both images below. *Is the bus to the left or to the right of the shelter in the picture?* The shelter remains at the same position within each image. The bus has the same shape, color, and height and is translated horizontally. The sky and road are identical. Left and right refer to the viewer's frame.

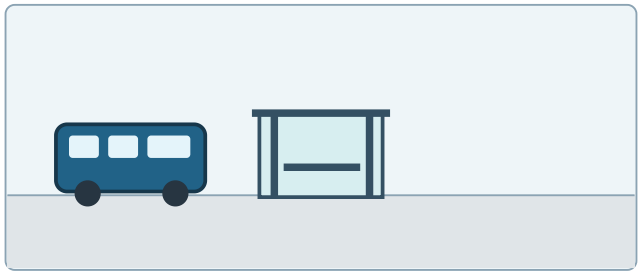


Image A

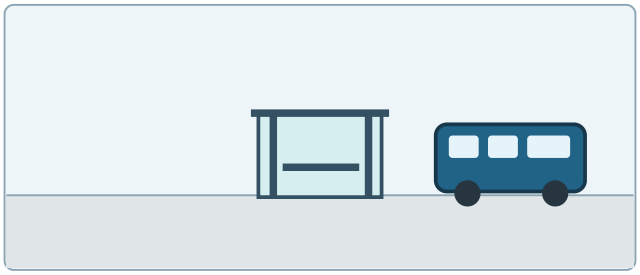


Image B

The expected answers are *left* for A and *right* for B. Crop each panel to the scene and rasterize it so that the input contains only pixels, with no panel label or SVG description. Send the two images in independent calls with the same question. For a larger test, build many such pairs with different scenes and buses. Check that the objects are visible and their horizontal positions are unambiguous. Balance the labels and randomize the presentation order. Keep both images in a pair in the same split. Use fixed generation settings and decide in advance how to map answers such as *to the left of it* to the label *left*.

Score each answer against its image and give a pair one point only if both answers are correct. Report the fraction of correct images and the fraction of fully correct pairs. A model that says *left* for both gets one image right and zero pair credit. A model that says *right* for A and *left* for B also gets zero pair credit. The answer must change in the correct direction.

A question-only baseline receives the exact same text for A and B. With deterministic evaluation and no image, it returns the same label both times. A fixed left or right answer yields 50 percent image accuracy and zero pair accuracy. An unchanged answer from the image model shows that the output failed to track the relation on that pair. Compare against the question-only baseline across pairs to test whether the image contributes useful information. An individual failure can also arise from confusing the objects or the words *left* and *right*, so it does not identify the model's internal cause.

11. Separate ambiguity from a wrong answer (optional)

Ask *What color is the bus's body?* on every image. Use photos or drawings where the body colors are clear, the buses are recognizable, and the windows and lights are easy to distinguish from the body. Annotate the number of buses, each body's accepted color names, and a usable location description for each bus before scoring responses. For example, record *blue bus on the left* and *red bus on the right* for a two-bus image.

Image type	Acceptable responses
One blue bus	*Blue* or *The bus is blue*. Accept agreed equivalents such as *navy blue* when the annotation supports them.
Blue bus on the left and red bus on the right	*Which bus do you mean?* or an answer that acknowledges both, such as *The left bus is blue and the right bus is red*.
No bus	*There is no bus visible in the image*.

A two-bus image leaves *the bus* unresolved. A response that asks which bus the user means is useful, and it makes no unsupported color claim. A bare *blue* is a visible color in the example but chooses one possible bus without justification. Score it as a failure to handle ambiguity. *The right bus is blue* is a different failure. Blue appears somewhere in the image, but the stated referent is red. On a no-bus image, assigning a bus a color also assumes a bus exists. On a single-bus image, asking which bus the user means is an unnecessary clarification.

Use two judgments for every output. First decide whether the response is acceptable under the table. Independently flag any claim about a bus color or its association with a particular bus that the image does not support. In a two-bus image, *Is it the green bus you mean?* sounds like a clarification but introduces a green bus absent from the image. Flag the unsupported claim and withhold credit for a grounded clarification. An output can fail the first judgment and also carry the unsupported claim flag.

Report appropriate-response rates separately for one bus, multiple buses, and no bus. Within the multiple-bus group, count how often the system asks a grounded clarification, states both valid possibilities, or chooses one without grounding the referent. Report the fraction of outputs with an unsupported color claim for each group, with raw counts. This keeps a legitimate request for clarification from being grouped with an incorrect guess. Use the same image resolution, prompt, and decoding configuration across systems and review borderline color terms under the same annotation rules.